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Course: CSE LAB

Course Code: 1202

Lab task Topic: Structured Programming Problems Solve {Array}

Submitted to:

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Problem No. 01

Name of the problem:

Write a program in C to store elements in an array and print it.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//print array number
#include<stdio.h>
int main()
{
   int a[4],i;
   printf("Enter 5 numbers\n");

   for(i=1;i<=5;i++)
   scanf("%d",&a[i]);

   for(i=1;i<=5;i++)
   printf("a[%d]=%d\n",i,a[i]);

   return 0;



}
```

Output:

```
"E:\Programming\my files\C++\array\array_1.exe"
Enter 5 numbers
1 2 3 4 5
a[1]=1
a[2]=2
a[3]=3
a[4]=4
a[5]=5

Process returned 0 (0x0)   execution time : 3.
786 s
Press any key to continue.
```

Problem No. 02

Name of the problem:

Write a program in C to read n number of values in an array and display it in reverse order

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//print reverse array numbers
#include<stdio.h>
int main()
{
   int a[5],i;
   printf("\tEnter 5 numbers\n");

   for(i=1;i<=5;i++)
            scanf("\t%d",&a[i]);

      printf("\tOutput\n");

   for(i=5;i>=1;i--)
   printf("\ta[%d]=%d\n",i,a[i]);

   return 0;

}
```

Output:

```
"E:\Programming\my files\C++\array\array_2.exe"
        Enter 5 numbers
1
2
3
4
5
        Output
        a[5]=5
        a[4]=4
        a[3]=3
        a[2]=2
        a[1]=1

Process returned 0 (0x0)   execution time : 4.
609 s
Press any key to continue.
```

Problem No. 03

Name of the problem:

Write a program in C to find the sum of all elements of the array.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Summation of array numbers
#include<stdio.h>
int main()
{
   int a[5],i,sum=0;
   printf("Enter 5 numbers\n");

   for(i=0;i<=4;i++)
   scanf("%d",&a[i]);

   for(i=0;i<5;i++)
      sum=sum+a[i];

   printf("The summation is: %d",sum);

   return 0;

}
```

Output:

```
"E:\Programming\my files\C++\array\array_3.exe"
Enter 5 numbers
1
2
3
4
5
The summation is: 15
Process returned 0 (0x0)   execution time : 3.
744 s
Press any key to continue.
```

Problem No. 04

Name of the problem:

Write a program in C to find the maximum and minimum element in an array.

Introduction

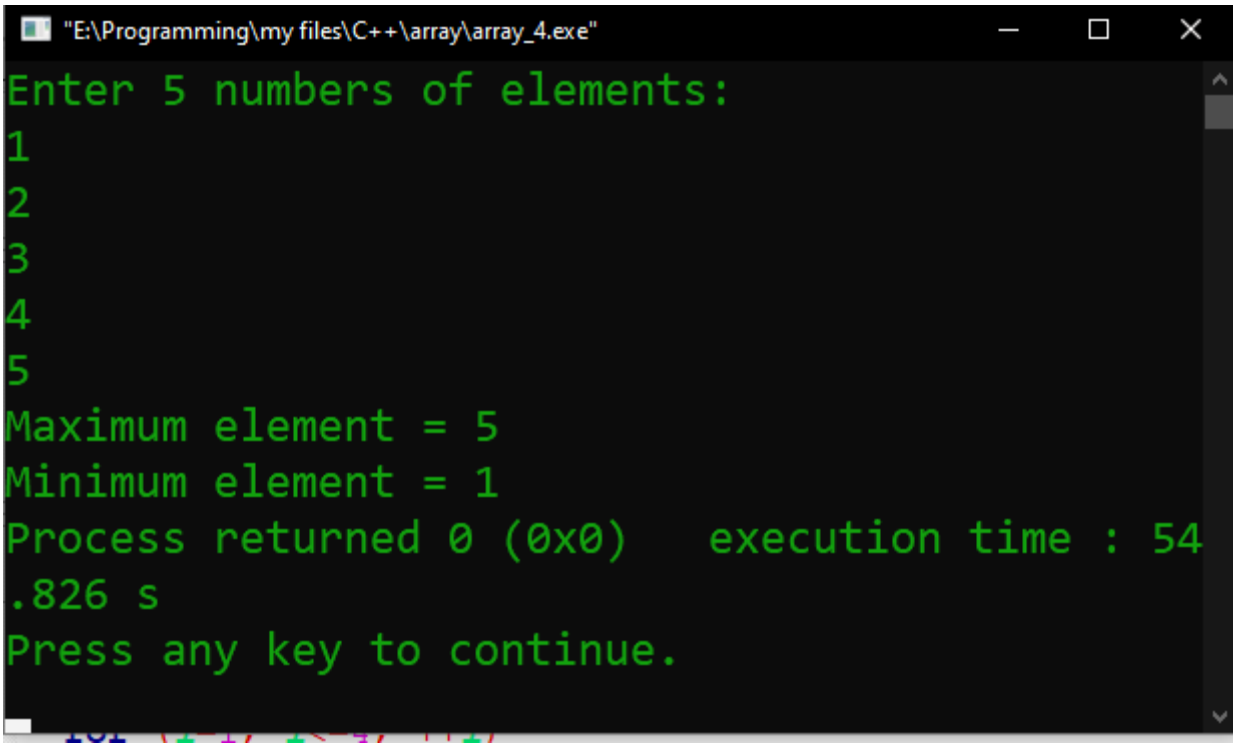
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//find Max & min value from array numbers
#include <stdio.h>
int main()
{
 int i,m,s;
 int a[5];
 printf("Enter 5 numbers of elements:\n");
 for (i=0; i<=4; ++i)
  {
  scanf("%d",&a[i]);
  }
  m=a[0];
  s=a[0];
 for (i=1; i<=4; ++i)
  {
  if (a[0]<a[i])
   {m=a[i];}
  if (a[0]>a[i])
   {s=a[i];}
```

```
  }
  printf("Maximum element = %d\n",m);
  printf("Minimum element = %d",s);
  return 0;
}
```

Output:



Problem No. 05

Name of the problem:

Write a program in C to sort elements of array in ascending order

Introduction

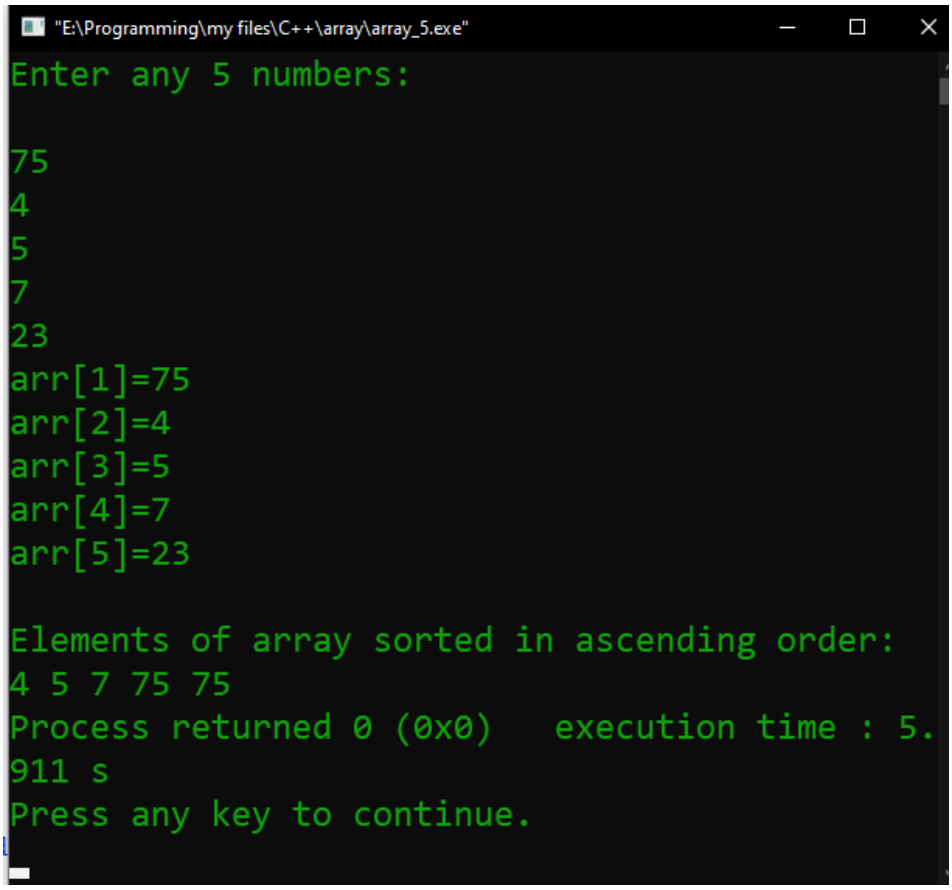
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//sort elements of array in ascending order
#include <stdio.h>
int main()
{
   int i,arr[5],j;
   int temp=0;
   printf("Enter any 5 numbers:\n");
   for(i=1;i<=5;i++)
   scanf("%d",&arr[i]);
   for(i=1;i<=5;i++)
   printf("arr[%d]=%d\n",i,arr[i]);
 //Sort the array in ascending order
   for (i=1; i<=5; i++)
   {
      for (j=i+1; j<=5; j++)
      {
        if(arr[i] > arr[j])
         {
            temp = arr[i];
```

```
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }
  }
  printf("\n");
  //Displaying elements of array after sorting
  printf("Elements of array sorted in ascending order: \n");
  for (i=1; i<=5; i++)
  {
    printf("%d ", arr[i]);
  }
  return 0;
}
```

Output:



Problem No. 06

Name of the problem:

Write a program in C program to Insert an Element in an Array.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Insert an Element in an Array
#include<stdio.h>
int main()
{
   char a[0],i;
   printf("Enter an element\n");
   scanf("%c",&a[0]);
   printf("Entered element is %c",a[0]);
   return 0;



}
```

Output:

```
"E:\Programming\my files\C++\array\array_6.exe"
Enter an element
1
Entered element is 1
Process returned 0 (0x0)   execution time : 4.
253 s
Press any key to continue.
```

Problem No. 07

Name of the problem:

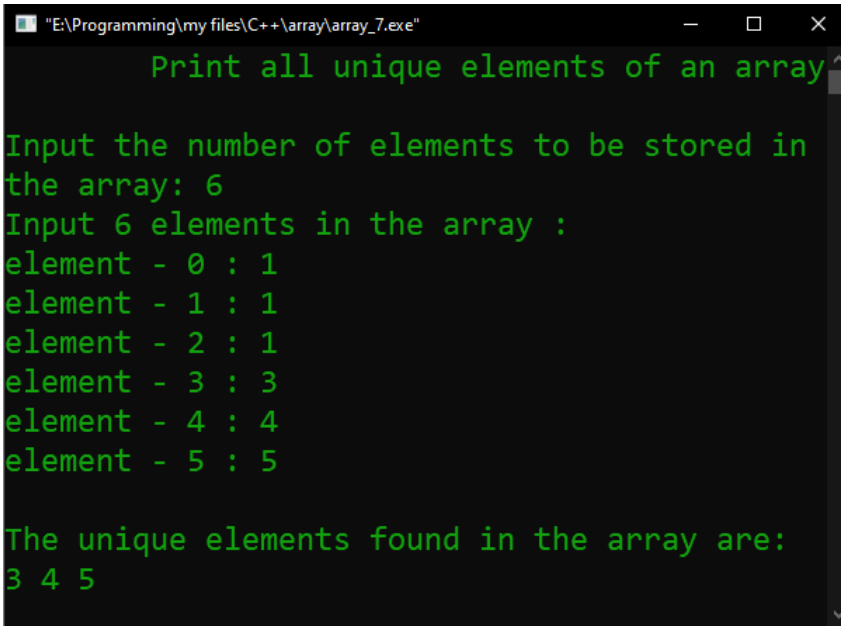
Write a program in C to print all unique elements in an array

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Print all unique elements of an array
#include <stdio.h>
int main()
{
   int a[100],n,ctr=0;
   int i,j,k;
     printf("\tPrint all unique elements of an array\n");
     printf("Input the number of elements to be stored in the array: ");
     scanf("%d",&n);
     printf("Input %d elements in the array :\n",n);
     for(i=0;i<n;i++)
      {
                printf("element - %d : ",i);
                scanf("%d",&a[i]);
      }
   printf("\nThe unique elements found in the array are:\n");
   for(i=0; i<n; i++)
   {
      ctr=0;
```

```
    for(j=0,k=n; j<k+1; j++)
    {
        /*Increment the counter when the search value is duplicate.*/
        if (i!=j)
        {
                    if(a[i]==a[j])
         {
            ctr++;
          }
         }
    }
   if(ctr==0)
    {
      printf("%d ",a[i]);
    }
  }
   printf("\n\n");
  return 0;
}
```

Output: